

SSI2100



PROCIRCUIT™ 512-STAGE BUCKET BRIGADE DELAY*

The SSI2100 is a new generation bucket brigade delay, developed on a modern IC process for performance improvements and new features while preserving the “mojo” of such beloved and quirky analog delay circuits.

A 512-stage bucket chain allows a wide range of delay times from a clock frequency of 1kHz to over 2MHz.

A novel circuit* allows easy connection of multiple SSI2100's for longer delay times without having to recalibrate inputs and outputs. An additional benefit to daisy-chaining is access to intermediate feedback taps for interesting reverb and other psychoacoustic effects.

An integrated clock driver greatly simplifies external clock sourcing and is TTL-compatible, making a companion BBD clock IC unnecessary.

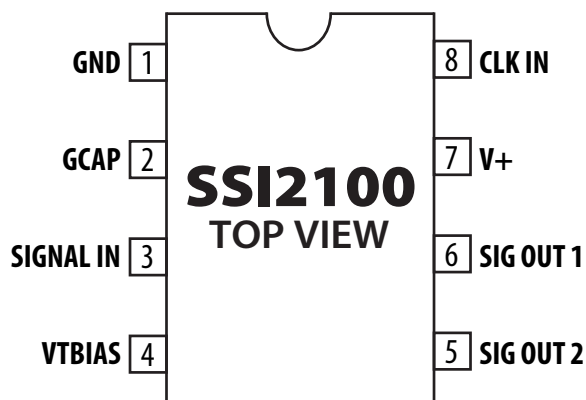
Also on-chip is a tetrode bias voltage generator that eliminates the need for externally providing the legacy “14/15 V_{GG}” supply voltage.

Refer to the Applications section for a high-performance Com-pandor circuit that significantly improves both SNR and input signal handling range.

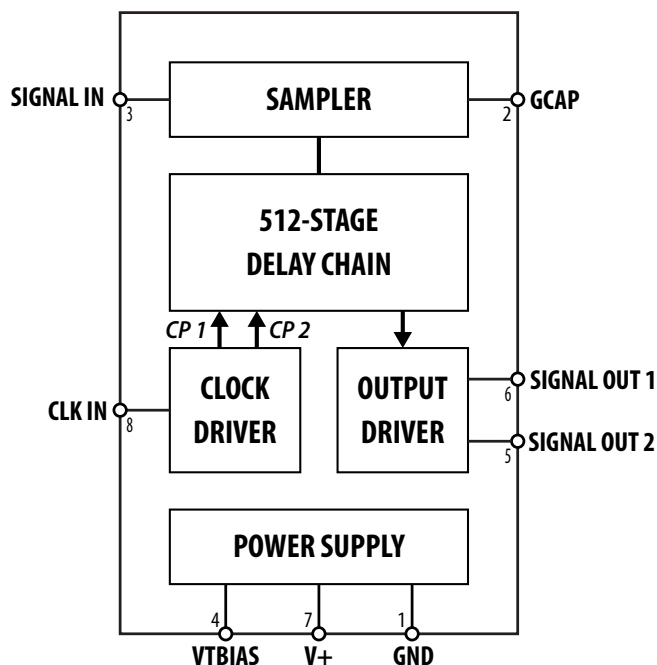
The SSI2100 is available in a space-saving 8-lead SOP sur-face-mount package.

FEATURES

- First Modern BBD Design in Decades
- 512-Stage Delay Chain With Clock Range from 1kHz to over 2MHz
- Easy Daisy-Chaining of Multiple Units for Longer Delays and Access to Intermediate Taps
- On-Chip Clock Driver with TTL/CMOS 5V and 3.3V Compatible Input
- 70dB Signal-to-Noise Ratio, which improves to 108dB using SSI Com-pandor Application ($f_{CLK} = 250kHz$)
- 0.40% THD+N ($V_{IN} = 20mV_{RMS}$)
- Compact SOP8 Surface-Mount Package
- Single 5V Operation



PIN CONNECTIONS
8-LEAD SOP



FUNCTIONAL BLOCK
DIAGRAM

*US Patent 12,652,025

The SSI2100 is available exclusively from Sound Semiconductor and its authorized resellers

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Rev. 2.5, June 2026

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SPECIFICATIONS ($V_+ = 5.0V$, $V_{DCB} = 3.2V$, $GCAP = GND$, $f_{CLK} = 40kHz$, $V_{IN} = 20mV_{RMS}$, $f_{IN} = 1kHz$, $T_A = 25^\circ C$; using Figure 1 circuit)

Parameter	Symbol	Conditions	Min	Typ	Max	Units
POWER SUPPLY						
Supply Voltage Range	V_+		4.75	5.0	5.25	V
Supply Current	I_{SY}			0.4	0.6	mA
Tetrode Bias Voltage Range	V_{TBIAS}		4.55	4.65	4.75	V
SIGNAL INPUT						
Frequency Response		Output 3dB down	20		15,000	Hz
Optimal Input Signal Range		THD = 2.5%, Compander Out	2.2		120	mV_{RMS}
		THD = 2.5%, Compander In ¹	0.002		3.25	V_{RMS}
Input Signal Dynamic Range		THD = 2.5%, Compander Out		35		dB
		THD = 2.5%, Compander In		64		dB
DC Bias Voltage	V_{DCB}	at Pin 3. See graph on Page 16		3.20		V
SIGNAL OUTPUT						
Output Offset Voltage		$GCAP = GND$	1.2	2.4		V
Signal Path Gain		$GCAP = V_+$		1.6	2.0	dB
				6.0		dB
CLOCK DRIVER INPUT						
Operating Frequency	f_{CLK}		1		2000	kHz
Logic Clock H Level	V_{IH}		2.5			V
Logic Clock L Level	V_{IL}				1.5	V
Logic Clock Rise/Fall Time	t_R, t_F				100	ns
Duty Cycle				50		%
Input Capacitance				5		pF
PERFORMANCE						
Output Noise Voltage		$V_{IN} = GND$, A-Weighted, $f_{CLK} = 250kHz$		38		μV_{RMS}
				-86		dBu
		Above, Compander In		12		μV_{RMS}
				-96		dBu
Signal-to-Noise Ratio	SNR^2	A-Weighted, 2.5% THD, $f_{CLK} = 250kHz$		70		dB
		Above, Compander In		108		dB
Total Harmonic Distortion	THD	Compander Out		0.4		%
		Compander In		0.56		%

¹See Page 10

² SNR is measured from the 2.5% THD point (120 mV_{RMS}) to the Output Noise Voltage. See graphs on Page 16 for further data.

ABSOLUTE MAXIMUM RATINGS

Supply Voltage	7V
Maximum Voltage into Pins	7V
Maximum Output Voltage	7V
Storage Temperature Range	-65°C to +150°C
Operating Temperature Range	-40°C to +85°C
Lead Temperature (Soldering, 10 sec)	260°C

ORDERING INFORMATION

Part Number	Package Type/Container	Quantity
SSI2100S-TU	8-Lead SOP* - Tube	100
SSI2100S-RT	8-Lead SOP* - Tape and Reel	4000

*SSI Package ID "PSL8", compliant with JEDEC MS-012-AA
Mechanical drawing available at www.soundsemiconductor.com

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PIN DESCRIPTIONS

Pin(s)	Name	Description
1	GND	Connect to analog signal ground with short, low inductance trace.
2	GCAP	Gain Capacitor Enable. Tie to GND for low gain, and to V+ for high gain when daisy-chaining multiple SSI2100s.
3	SIGNAL IN	Audio signal input. Requires a DC bias of 3.20V for correct operation, and a low-impedance source to drive the internal sample-and-hold.
4	VTBIAS	Internally-generated 14/15 of V+. Connect a 3.3μF capacitor to GND.
5	SIG OUT 2	Output when CLK IN is high. Source follower.
6	SIG OUT 1	Output when CLK IN is low. Source follower.
7	V+	+5V supply. Recommend 100nF local decoupling capacitor placed as close to package as possible with a low inductance trace to ground.
8	CLK IN	Clock input to internal clock generator, producing two anti-phase buffered clock signals for the delay chain. Compatible with both 3.3V and 5V logic thresholds.

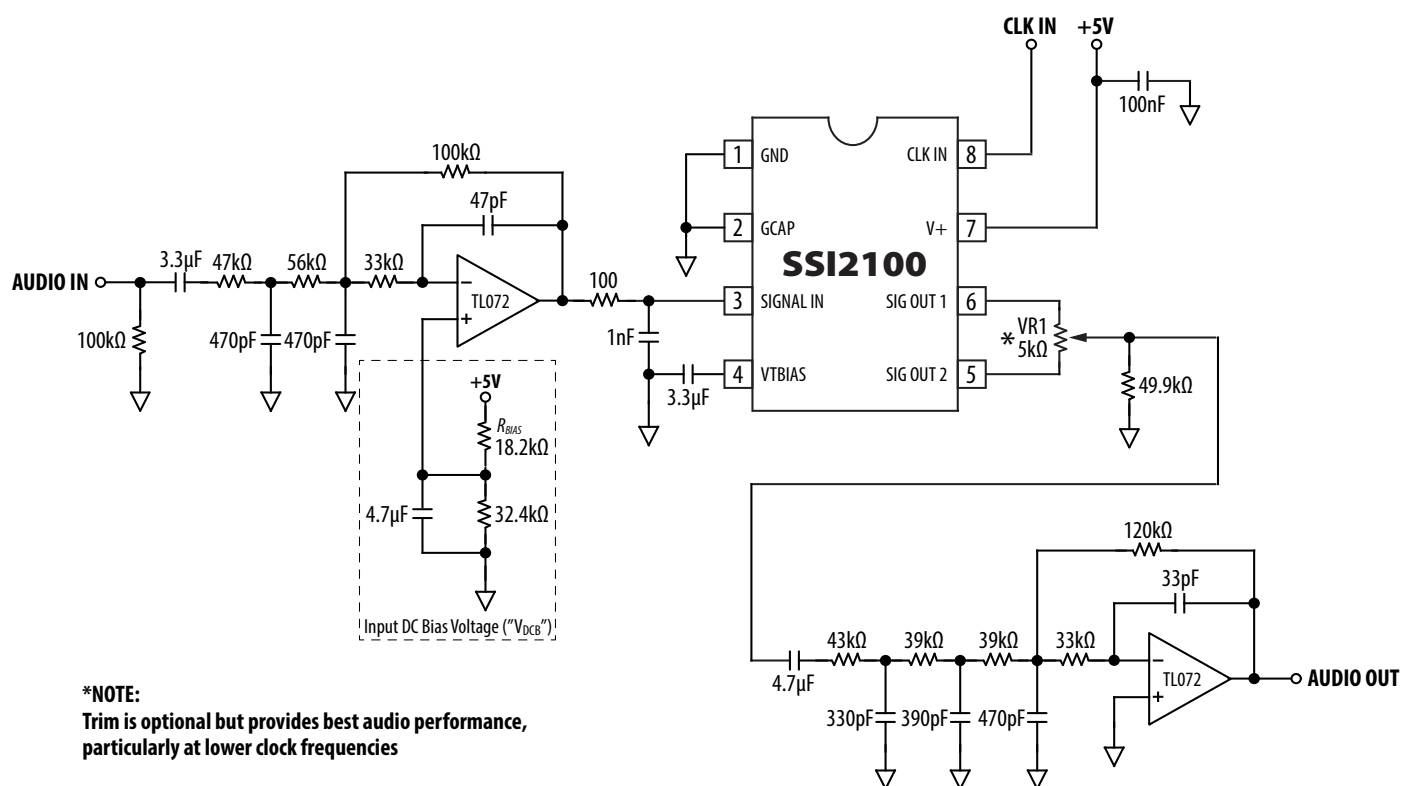


Figure 1: Typical Application Circuit
Input and output filters are set to 15kHz cut-off frequencies for $f_{CLK} \geq 30\text{kHz}$.

USING THE SSI2100

The SSI2100 is a new generation CMOS bucket brigade delay (BBD). It comprises a sample-and-hold with programmable storage capacitor, a 512-stage switched capacitor delay line, a two-stage output driver, internal clock driver and clock tree, and internal V_{TBIAS} network.

Typical Application

Figure 1 shows a typical application circuit. BBDs are sampled data devices so the first circuit block is an anti-aliasing filter on the input to remove any signal frequencies above half the clock frequency. The filter as shown is a 3-pole low-pass filter with AC-coupled input, and the

DC bias applied to the filter as well. The filter output couples to the input of the SSI2100 through an RC network to stabilize the signal during sampling windows. In this example the gain setting pin GCAP is connected to ground to give a signal path gain of about 1.6 dB.

The output of the SSI2100 is a train of pulses from the two source-follower outputs. The outputs are combined into a common load resistor, and then AC-coupled into a 4-pole low-pass reconstruction filter, again set to half the clock frequency.

The internal V_{BIAS} network needs an external 3.3 μF capacitor connected from pin 4 to ground.

The SSI2100 requires a single TTL/CMOS-compatible square-wave clock signal to drive the internal clock tree.

Power Supply and Grounding

The SSI2100 is a 5V-only device. The positive supply $V+$ must be fed from a regulated 5V source. It should have a local 100nF good quality ceramic decoupling capacitor placed close to the supply pin, together with local bulk supply rail decoupling capacitors of 10 μF or more, and with good ground and supply trace routing practices applied throughout.

The SSI2100 has one ground connection on pin 1. It must be connected to ground using thick short traces or closely positioned vias to a ground plane.

Signal Input

The signal input (pin 3) to the SSI2100 looks like a sample-and-hold operating at the clock frequency. When the clock signal is low the sampling switch is open and connects the signal source to the internal sampling capacitor. The sampling capacitor is between 5pF and 10pF depending on the gain setting (see below).

Under certain signal conditions a considerable short pulse of current can occur at the start of the sampling period. It is recommended that an external 1nF capacitor and series 100 Ω resistor are used to isolate op amp outputs from this dynamic load, which could otherwise cause the op amp to ring at each sample point.

For most applications, a compandor is recommended for significant improvements in both input signal handling and signal-to-noise ratio. A legacy compandor can be used, but for best results the SSI2162-based compandor (see Page 10) improves maximum signal handling from 120mV_{RMS} to 3.25V (2.5% THD) and SNR from 70dB to 108dB.

The signal input also needs a DC bias (" V_{DCB} ") of 3.20V – there are several ways to achieve this; the circuit in Figure 1 applies the bias as an offset to the input filter. Care should be taken in the design of the bias system to ensure it has a low output impedance.

BBDs are inherently sampled-data systems. As such the applied signal must meet the Nyquist criterion of no frequency components greater than the clock frequency; otherwise they will alias down into the audio band and introduce unwanted distortion. To prevent this it is recommended to use an anti-alias filter, such as shown in Figure 1, which is a typical 3-pole multi-feedback low-pass filter. The component values shown set the cutoff frequency at 15kHz. To change to a different cutoff frequency the easiest approach is to scale the capacitors:

$$C_{\text{NEW}} = C_{\text{OLD}} \frac{15,000}{f_{\text{NEW}}}$$

The small R-C network separates the sampling input of the BBD from the opamp output – this helps keep the op amp stable during the initial part of the sampling window when the internal sampling capacitor charges up from the external signal.

For applications with variable clock frequencies you can either implement a sweepable filter to track the clock frequency, or use a fixed filter set to the highest frequency of interest and then ensure that the minimum clock frequency is at least twice that.

Gain Control (GCAP)

The gain capacitor selection input (pin 2) enables and disables an additional internal capacitor in the sample-and-hold circuit. Due to the way BBDs operate this has the effect of applying additional gain to the signal at the front of the BBD for applications that may require it, such as chaining (discussed later).

The GCAP pin must be tied either to GND for standard gain, or to $V+$ for high gain. It must not be left floating.

Signal Outputs

The SSI2100 has two antiphase outputs from the BBD chain. Each output is a source-follower from an internal n-channel MOSFET. External circuits must provide a load to the outputs, either individually or commoned. Outputs may be used independently as well, such as when chaining.

Both outputs contain considerable clock energy which must be removed using a reconstruction filter. This filter has the same Nyquist frequency requirement of the input filter. The output filter shown in Figure 1 is a 4-pole low-pass filter with AC-coupled input to remove the DC output bias. For different cutoff frequencies the same capacitor scaling used for the anti-alias filter can be used.

Clocking

The clock input (pin 8) is compatible with TTL and CMOS logic levels in both 5V and 3.3V systems. It should be driven by a square wave with equal high and low periods and fast transitions. If your design cannot guarantee this requirement then we recommend using a flip-flop to ensure a clean square wave, and then to double the frequency range of the clock source to compensate.

The clock frequency range of the SSI2100 is very wide, and will perform within specifications from 1 kHz to over 2MHz. It can be clocked both lower and higher with some degradation in performance. As the SSI2100 is a sample-based device its behavior is subject to Nyquist limits, where the clock frequency needs to be at least twice the highest frequency present in the input signal.

Tetrode Bias (V_{TBIAS})

The SSI2100 has an internal V_{TBIAS} generator. For normal operation it requires a 3.3 μF capacitor connected from pin 4 to ground. This voltage (sometimes called V_{GG}) is used to bias the internal tetrode transistors in the delay chain to improve performance. Refer to the Principles of Operation section for a description of the tetrode transistors.

Critical Component Selection

There are very few critical components for the SSI2100.

The supply decoupling capacitor should be a good quality 100nF ceramic capacitor of at least X7R dielectric. It should be placed as close to the supply pin as possible, with a low-impedance trace to ground.

The V_{TBias} capacitor should be a good quality electrolytic. For very high clock frequencies it is recommended to connect a smaller value capacitor (e.g., 100nF ceramic X7R) in parallel to reduce impedance of the bypass capacitance.

Daisy-Chaining Multiple SSI2100s

The chaining circuit is shown in Figure 2 (only the chaining components are shown, all other components are the same as for non-chain applications). Sound Semiconductor has not yet confirmed chained operation beyond four chips; users who have experimented with more chips are encouraged to share their findings.

In this circuit only the OUT1 signal is required since it will be resampled by the input of the following BBD. The other output, OUT2, is available for other uses such as feeding some of the intermediate signal back to the input after it has been processed through a reconstruction filter.

Resistor R1 is the load for the output transistor, while R2 and R3 simultaneously bias and scale the signal to match the input of the following SSI2100. Capacitor C1 reduces the transient impedance of the resistor network to drive the sampler of the following BBD.

The recommended clocking scheme is to drive all SSI2100 from the same clock source as shown. This ensures that OUT1 is active at the correct time to be sampled by the following BBD.

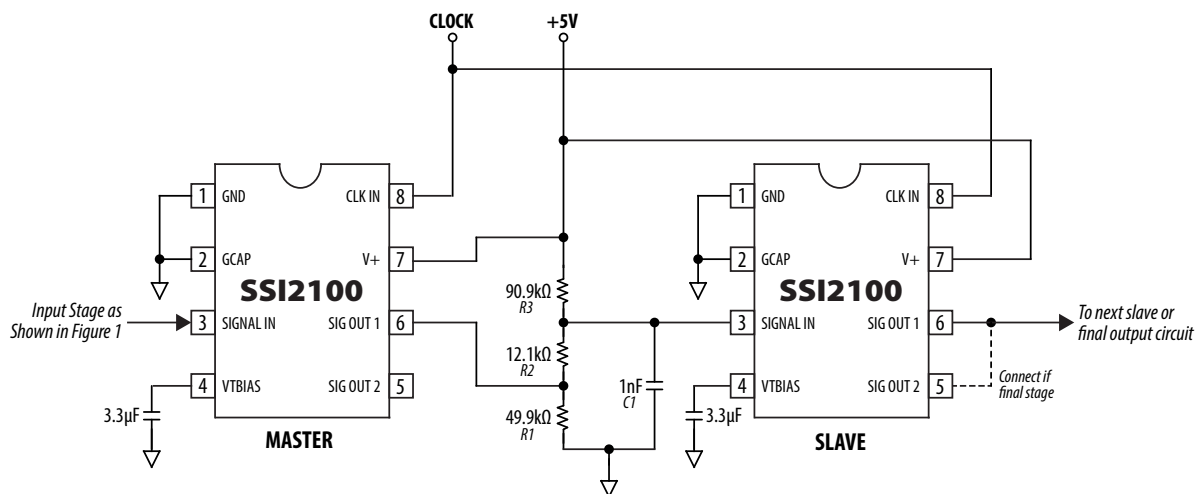


Figure 2: Daisy-Chaining Scheme

Design Hints for Best Audio Performance

In addition to the aforementioned compandor circuit on Page 10, the designer can take certain precautions to achieve optimal performance.

The two output pins provide anti-phase outputs that provide a good signal level without an external sample-and-hold circuit. However, mismatches in the output transistors result in small output level differences that can allow some clock noise leakage. To compensate for this a trimmer is used as shown in Figure 1.

Trimming the output can be either by ear or measurement (e.g., oscilloscope or audio analyzer). Start with the trimmer at one end of the track and adjust until the minimum clock noise point is found - you might need to move back and forth to find the best trim point. Once set it is recommended to lock the trimmer with a drop of paint or similar (nail varnish is rather good at this).

In applications where the clock signal is sufficiently high and the reconstruction filter transition band is steep enough, the trim circuit may not be required (see Figure 3). This circuit simplifies the output stage and avoids a production trim but is only suitable when the clock frequency is much higher than the audio band; for example clock frequencies above 100 kHz and reconstruction filters with corner frequencies below 20kHz and 3-pole or better.

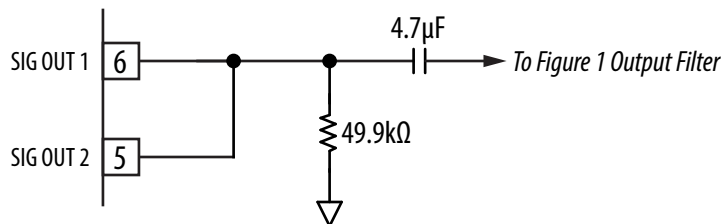


Figure 3: Untrimmed Output - only advised for $F_{clk} \gg 40$ kHz

In addition to correct selection of output trimming and reconstruction filter parameters, it is also important to pay attention to other parts of the signal path. The SSI2100 should be decoupled locally with a good quality 100nF ceramic capacitor. Some applications may also benefit from a much larger local bulk capacitor and low-value series resistor to limit the spread of high-frequency clock noise through the power rails. Use a good quality low-ESR aluminum electrolytic for best results.

Good PCB layout practices are essential in separating the input, output, and clock traces from each other. The clock signal - with its sharp edges and large amplitude - is particularly adept at capacitively coupling into other parts of the signal path. The impedance of the ground circuit must be kept as low as possible, and the circuit loop from the clock source to the SSI2100 and back must be kept as small as possible.

In practice, this means several specific layout recommendations. Route the clock trace as a direct, short path between the clock source and the SSI2100, avoiding long parallel runs alongside input or output signal traces. Where a clock trace must cross an audio trace, route them on different layers and cross at right angles to minimize capacitive coupling. Guard traces (grounded copper running either side of the clock line) can further contain the field, though they are only effective when tied to a solid ground plane at frequent intervals such as 5mm or less.

An unbroken ground plane directly beneath the SSI2100 and its associated clock and signal traces is the single most effective measure. Avoid splits or cutouts in the ground plane under any high-frequency trace, as these force return currents to detour, enlarging the current loop and radiating accordingly. The return current for the clock signal will naturally flow in the ground plane directly below the clock trace, provided the plane is continuous; interrupting this path is a common source of otherwise-mysterious clock bleed.

Keep the decoupling capacitor's ground return as close to the SSI2100's ground pin as physically possible — ideally within 1-2 mm. The inductance of even a short trace between the capacitor and the IC can significantly degrade decoupling at clock harmonics. Where board space permits, place the decoupling capacitor on the underside of the board directly beneath the ground and supply pins.

Treat the clock oscillator itself as a noise source. Keep it physically separated from the audio input and output traces, and do not route audio signals beneath the oscillator or its bypass components.

The two outputs - SIG OUT 1 and SIG OUT 2 - also contain high frequencies which must be kept as close to the SSI2100 as possible and not allowed to couple into other parts of the signal path. The reconstruction filter should be placed next to the SSI2100 output pins. Only after the reconstruction filter should the audio signal be routed to other parts of the circuit.

PRINCIPLES OF OPERATION

The SSI2100 internal structure is shown in Figure 4. It comprises the input sampler, the 512-stage bucket brigade delay chain, the output driver, clock driver, and V_{TBIAS} generator. Each is described in detail in the following sections.

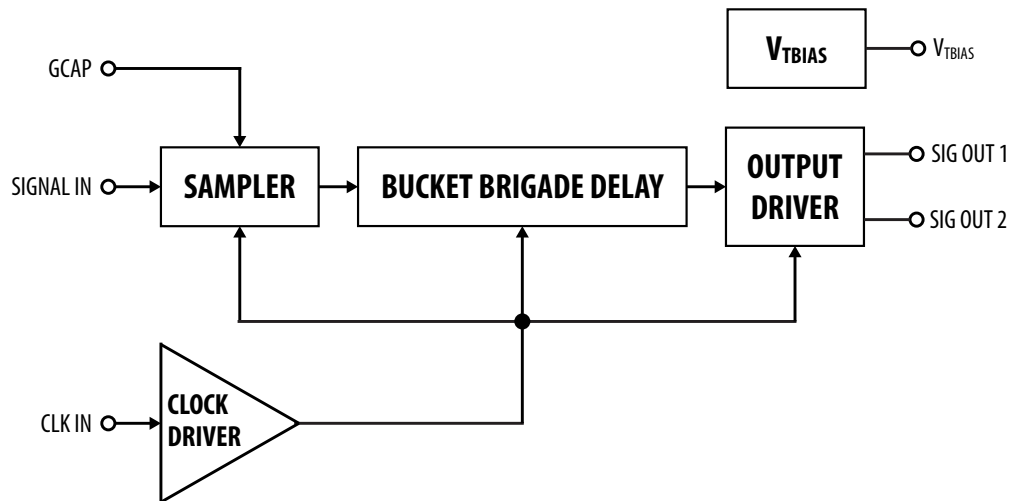


Figure 4: Simplified Internal Structure

Sampler

See Figure 5 for the first stage of the BBD is the sampler. The incoming signal is sampled when Q1 is turned on, and a charge proportional to the instantaneous signal level accumulates on sampling capacitor C1. When Q1 is turned off the charge on C1 is available to be transferred to the delay chain.

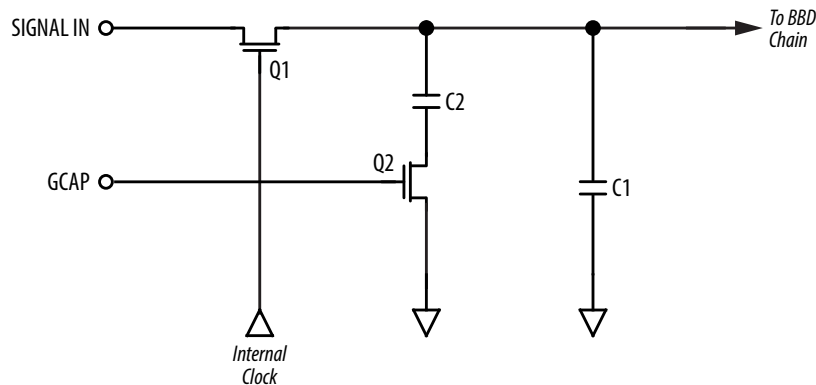


Figure 5: Sampler

Gain Control

The “gain” of a BBD is related to the ratio of the sampling capacitor (C1) and the delay capacitors (C3, C4 in Figure 6):

$$G = \frac{C_1}{C_3}$$

A novel* feature of the SSI2100 is an additional sampling capacitor, C2, that can be enabled by turning on Q2. This extra capacitance increases the signal gain through the BBD without introducing noise since there are no amplifiers involved.

Bucket Brigade

Figure 6 shows the bucket brigade structure. Each delay stage contains two delay capacitors (C3 and C4), two tetrode transistors (Q3 and Q5), and two switching transistors (Q4 and Q6).

The tetrode transistors form a cascode structure with the switching transistors, and gives two important benefits:

1. Improves overall transconductance (helps stabilize operating point for a better g_m).
2. Reduces capacitive coupling of clock pulses (Miller effect).

*US Patent 12,652,025

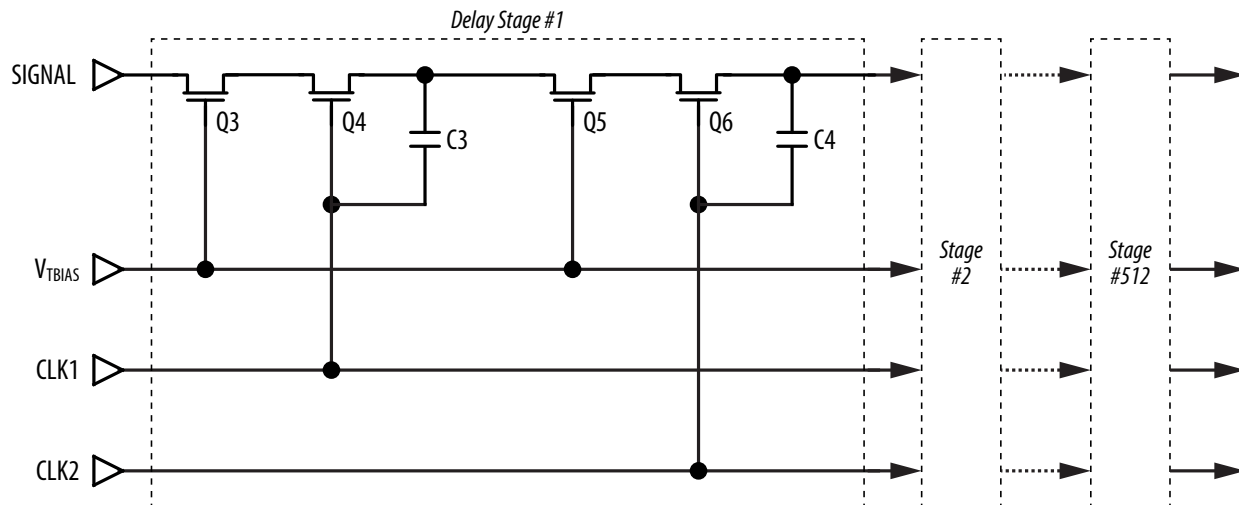


Figure 6: Bucket Brigade

The name “tetrode” comes from thermionic valve technology and simply means the device has four electrodes: drain, source, and two gates. In FET transistors they are commonly called dual-gate FETs and are widely used in RF applications.

The switching transistors, together with the bi-phase clocks (CLK1 and CLK2), alternately charge and discharge the delay capacitors. Because capacitors store charge in the form of negatively-charged electrons on their plates, the effect is to transfer a lack of charge from the input to the output.

This is often likened to a chain of buckets, from which the name is derived (see Figure 7): the bucket at the output is filled from a tap, and at each clock cycle alternate buckets empty themselves into the preceding bucket until that bucket is full; in the diagram below at the next clock cycle the B buckets will empty themselves to fill up the A buckets, the result of which is that the empty portions move to the right, from the input to the output.

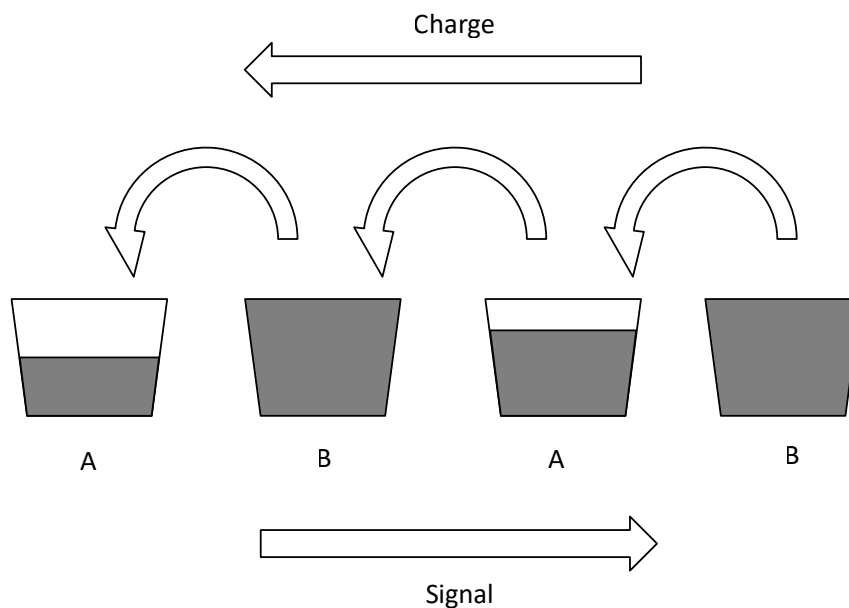


Figure 7: How a Bucket Brigade Works

The delay in transferring the signal from input to output is given by (in seconds):

$$T_{delay} = \frac{N_{stages}}{2F_{clk}}$$

The table below gives the delay times for a range of clock frequencies from the SSI2100's 512 stages:

Clock Frequency (kHz)	Delay (ms)
10	25.6
20	12.8
50	5.12
100	2.56
200	1.28
500	0.512
1000	0.256

Output

At the end of the delay chain is the output stage. It is similar to a single delay stage but with the addition of two source-follower output drivers (Q13 and Q14). See Figure 8.

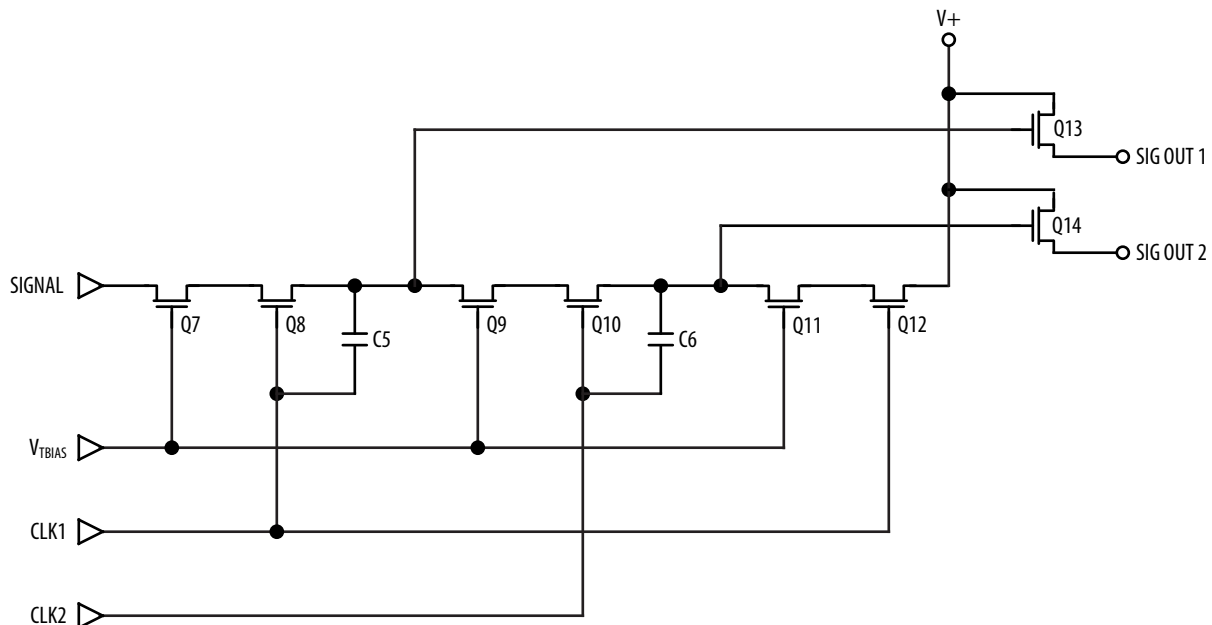


Figure 8: Output Stage

Each output reflects the voltage (charge) on each of the two capacitors C5 and C6. When the capacitor is fully charged the output will be at the highest voltage, and when the capacitor holds a sample of the input signal the output will be lower than the maximum. In principle only one output is necessary to reconstruct the original signal (and as used when chaining) although this is best done with an external sample-and-hold and stiff reconstruction filter. However combining both outputs together greatly reduces clock noise, removes the need for an external sample-and-hold, and eases the job of the filter.

Clock Driver

The SSI2100 incorporates an internal clock driver and distributed clock tree as shown in Figure 9. This solves two issues of legacy devices: highly capacitive clock inputs requiring two high-current drivers, and care in ensuring the edges are clean.

The clock driver first brings the external clock signal with a low-capacitance input which handles both 3.3V and 5V logic levels. The next two stages buffer this signal and also derive the anti-phase clock signal.

The two internal clock signals CLK1 and CLK2 are then distributed along the delay chain with further clock drivers arranged in groups of delay stages. This simplifies the clock loading and ensures clean clock signals are delivered to all delay stages within the SSI2100.

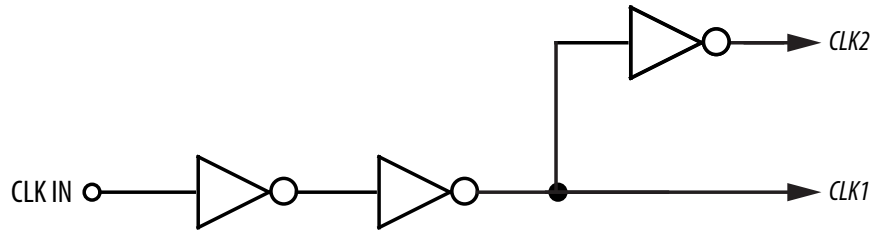


Figure 9: Clock Driver

V_{TBIAS}

The tetrode transistors require their gates to be biased at about 14/15th of $V+$. This voltage is generated internally, as shown in Figure 10, but requires an external capacitor; typically 3.3 μ F.

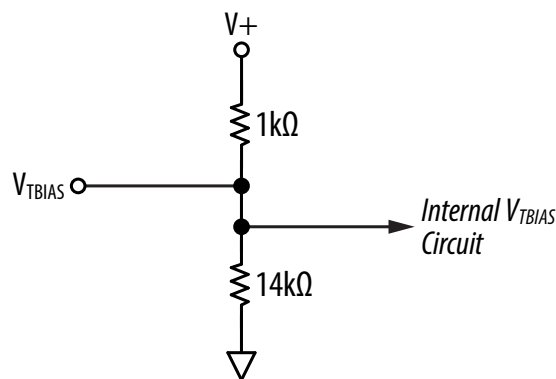


Figure 10: V_{TBIAS}

APPLICATION INFORMATION

A HIGH-PERFORMANCE COMPANDOR

BBDs have a relatively small useful signal range – around 40dB – which limits their signal-to-noise ratio. To improve performance, a “Compan-dor” (Compressor + Expander) is used to compress the input signal before the BBD, and then expand the output of the BBD back to the original signal range. Figure 11 shows the relationships between the three dynamic ranges (Input, BBD and Output).

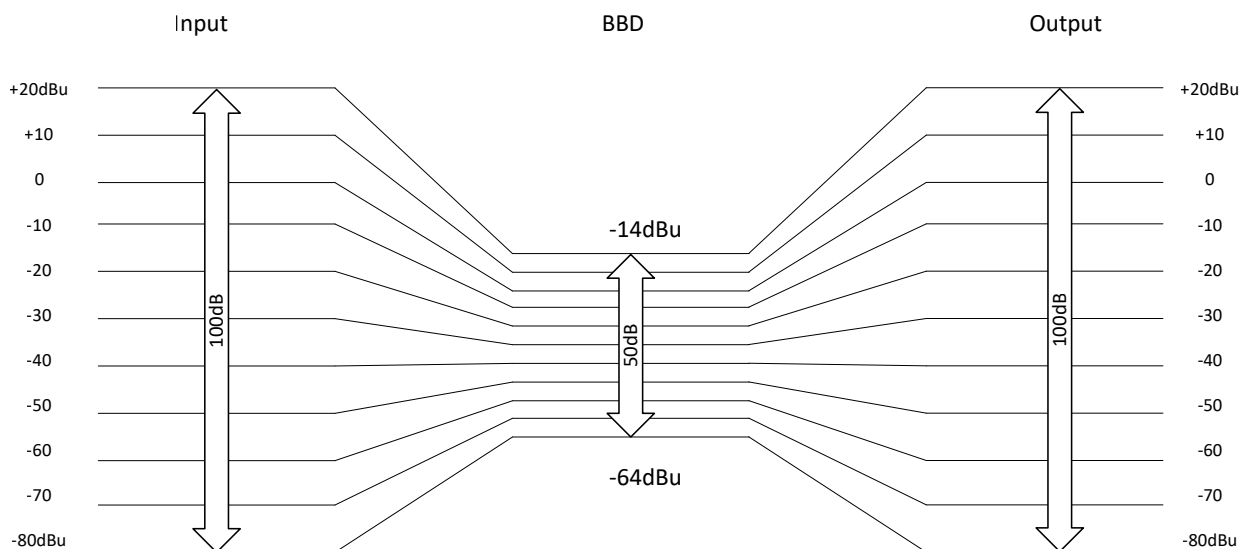


Figure 11: Basic Compandor Operation

Typical audio signals can range greatly from -80dBu to +20dBu. The compressor lifts and compresses the input to a much smaller -64dBu to -14dBu range that better suits the BBD's input range. After passing through the BBD, the signal is then expanded back to the original -80dBu to +20dBu range. The complementary halves of the compandor work together to achieve this, with dual benefits of increased signal-to-noise ratio and audio input signal handling range.

Compressor

The compressor circuit shown in Figure 12 is a feedback type, where the input to the sidechain processor comes from the compressor's output and feeds back to control gain. For signals below about -80dBu the signal path has a fixed gain of +16dB (x6.2 from the 10kΩ VCA input and 62kΩ op amp feedback resistor). This lifts the -80dBu signal up to -64dBu.

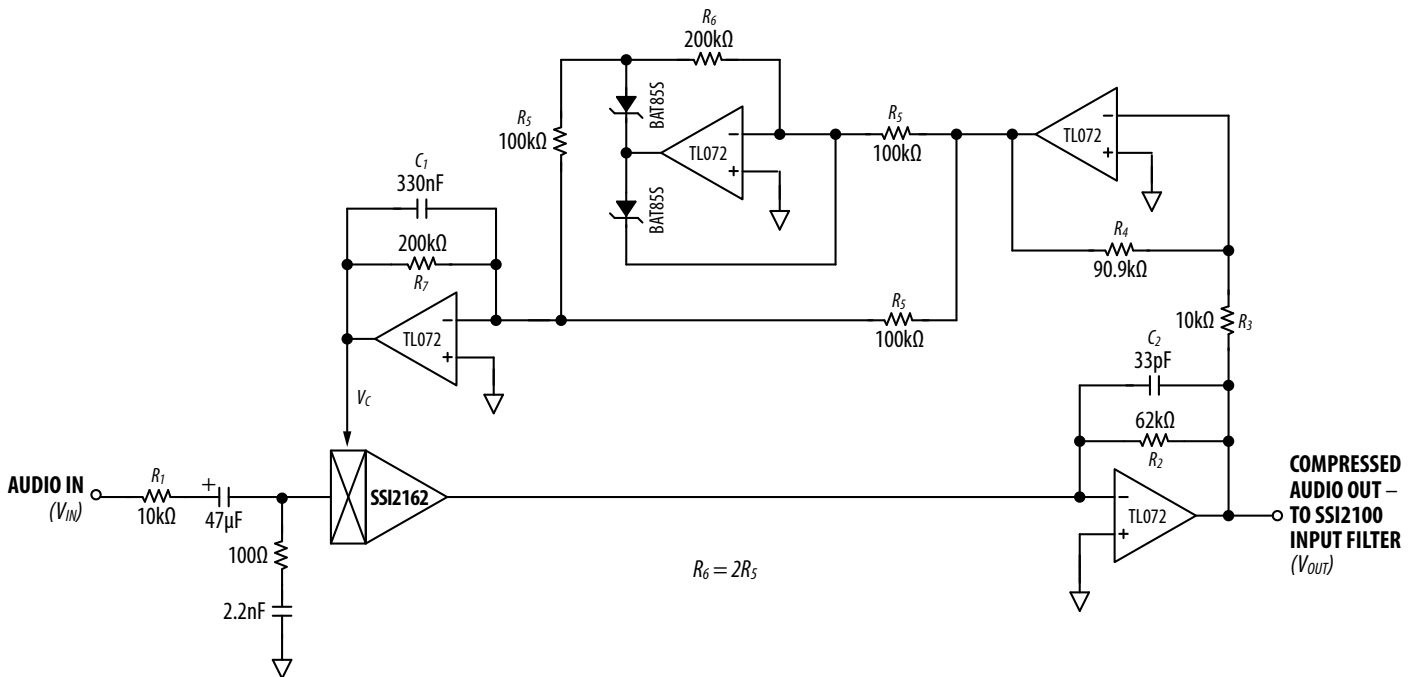


Figure 12: Compressor

The output signal V_{OUT} is passed on both to the BBD's input anti-alias filter ("AUDIO IN" on Figure 1) and to the input of the sidechain circuit. The signal is amplified before going into an inverting full-wave rectifier based around an op amp and two Schottky diodes. The resistors around the rectifier generate a fully-rectified and inverted current into the final op amp configured as a current-in voltage-out transimpedance amplifier. The current is then smoothed and converted to a positive voltage to control one half of an SSI2162 dual audio VCA resulting in reduction (compression) of the audio signal.

The behavior of the compressor is interesting to analyze, starting with the audio signal path. The output voltage, V_{OUT} , is:

$$V_{OUT} = V_{IN} \frac{R_2}{R_1} 10^{-\frac{3}{2}V_C} \quad (1)$$

where $\frac{R_2}{R_1}$ is the fixed gain of the op amp, and $10^{-\frac{3}{2}V_C}$ is the gain of the SSI2162 VCA. The sidechain circuit is rather more complicated. Its input is V_{OUT} from which it generates V_C :

$$V_C = \frac{R_7}{R_5} \frac{R_4}{R_3} |V_{OUT}| \quad (2)$$

This particular circuit uses a peak detector whereas the signal levels are in RMS, so we need to convert from V_{PEAK} to V_{RMS} :

$$V_C = \frac{R_7}{R_5} \frac{R_4}{R_3} \frac{1}{\sqrt{2}} V_{OUT} \quad (3)$$

Substitute (3) into (1) and collect the terms:

$$V_{OUT} 10^{\left(\frac{3 R_7 R_4}{2 R_5 R_3} \frac{1}{\sqrt{2}} V_{OUT}\right)} = \frac{R_2}{R_1} V_{IN} \quad (4)$$

Unfortunately, the expression in V_{OUT} is a transcendental function: V_{OUT} is in both the multiplier and in the exponent term itself. This means it is impossible to solve just by re-arranging it. To solve this we need to use the Lambert W Function, which is the inverse of:

$$f(W) = We^W$$

Note: you won't find the Lambert function on your calculator, but you will in Matlab, Maxima, GNU Octave, Python, and many other computer-based math packages. For real values the zeroth Lambert W function looks very similar to the log function: a curve that rises and bends to the right like the letter "r".

To use the Lambert W function the exponent base must be scaled from 10 to e:

$$V_{OUT} e^{\left(\frac{3 R_7 R_4}{2 R_5 R_3} \frac{1}{\sqrt{2}} \ln(10) V_{OUT}\right)} = \frac{R_2}{R_1} V_{IN} \quad (5)$$

Now introduce a new term:

$$u = \frac{3 R_7 R_4}{2 R_5 R_3} \frac{1}{\sqrt{2}} \ln(10) V_{OUT} \quad (6)$$

so that

$$V_{OUT} = \frac{u}{\frac{3 R_7 R_4}{2 R_5 R_3} \frac{1}{\sqrt{2}} \ln(10)} \quad (7)$$

Substituting (6) and (7) into (5) gives

$$\frac{u}{\frac{3 R_7 R_4}{2 R_5 R_3} \frac{1}{\sqrt{2}} \ln(10)} e^u = \frac{R_2}{R_1} V_{IN} \quad (8)$$

Then re-arrange (8) into

$$ue^u = \frac{3 R_7 R_4 R_2}{2 R_5 R_3 R_1} \frac{1}{\sqrt{2}} \ln(10) V_{IN} \quad (9)$$

Take the Lambert of both sides:

$$W(ue^u) = W\left(\frac{3 R_7 R_4 R_2}{2 R_5 R_3 R_1} \frac{1}{\sqrt{2}} \ln(10) V_{IN}\right)$$

$$u = W\left(\frac{3 R_7 R_4 R_2}{2 R_5 R_3 R_1} \frac{1}{\sqrt{2}} \ln(10) V_{IN}\right) \quad (10)$$

Substitute back for u in (10):

$$\frac{3 R_7 R_4}{2 R_5 R_3} \frac{1}{\sqrt{2}} \ln(10) V_{OUT} = W\left(\frac{3 R_7 R_4 R_2}{2 R_5 R_3 R_1} \frac{1}{\sqrt{2}} \ln(10) V_{IN}\right) \quad (11)$$

Finally, re-arrange (11) to give V_{OUT} only in terms of V_{IN} :

$$V_{OUT} = \frac{W\left(\frac{3 R_7 R_4 R_2}{2 R_5 R_3 R_1} \frac{1}{\sqrt{2}} \ln(10) V_{IN}\right)}{\frac{3 R_7 R_4}{2 R_5 R_3} \frac{1}{\sqrt{2}} \ln(10)} \quad (12)$$

This is now in a form that can be used to design and adjust the compressor response. Using the component values from Figure 12 produces the expected compression transfer curve of Figure 13:

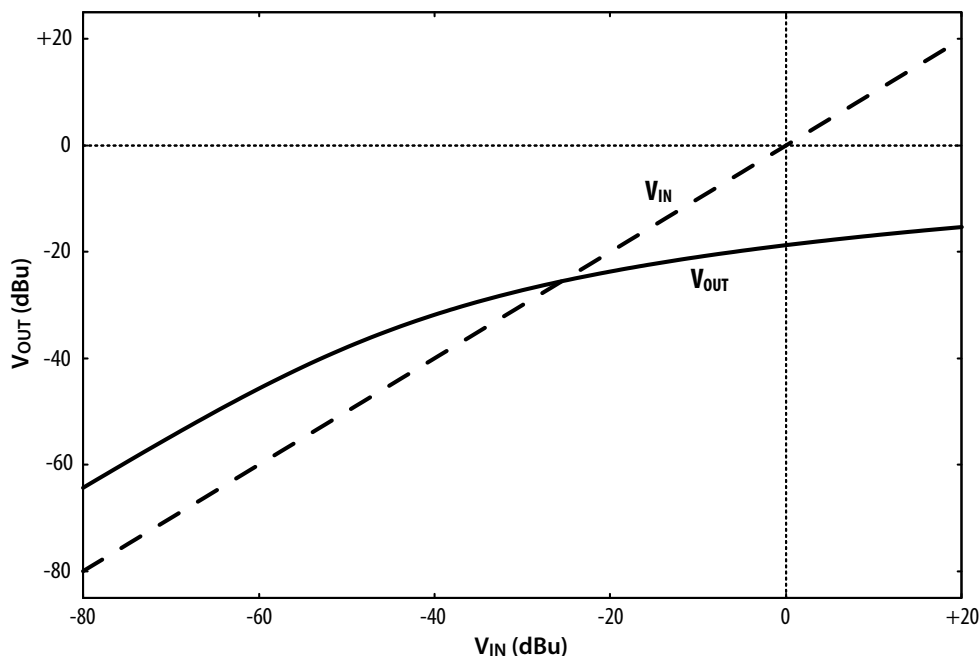


Figure 13: Compressor Transfer Curve

The dotted line plots the input signal and solid line the output signal. As expected, at low amplitudes the signal is boosted. At around -25dBu the compressor is at unity gain where the output line crosses the input line. By this stage the VCA is already reducing the signal level, and after this point the output rises much slower than the input. The compressor exhibits a very soft “knee” and the compression ratio gradually increases.

The other compressor aspect to consider is attack and release behavior. Attack time is of the order of 5-10ms depending on the audio material. Due to the gain of the sidechain, the attack sees a very strong drive signal which quickly ramps up the control voltage in about 1/10th of the decay time. This reduces the signal into the sidechain circuit and rapidly ends the attack.

Release time is set by the values of C1 and R7:

$$\tau_R = R_7 C_1$$

Component values shown result in about 66ms.

The key components in this circuit and their function are set out in the following table, together with typical values as shown in the schematic:

Component	Function	Value
R1	Sets the input resistance	10kΩ
R2	Sets the make-up gain	62kΩ
R3	Sets sidechain input resistance	10kΩ
R4	Sets pre-rectifier gain	90.9kΩ
R5	Sets rectifier gain	100kΩ
R6	= 2x R5	200kΩ
R7	Together with R5 sets the transimpedance amp, and with C1 sets the attack/release time constants	200kΩ
C1	With R7 sets the attack/release time constants	330nF
C2	Sets HF rolloff of the signal op amp	33pF

Expander

At the output of the BBD's filter is the expander (Figure 14). This has an equal but opposite gain response to the compressor. One significant difference to the compressor is that the expander's sidechain signal is taken from the input, not the output, and so it's of the feed-forward variety.

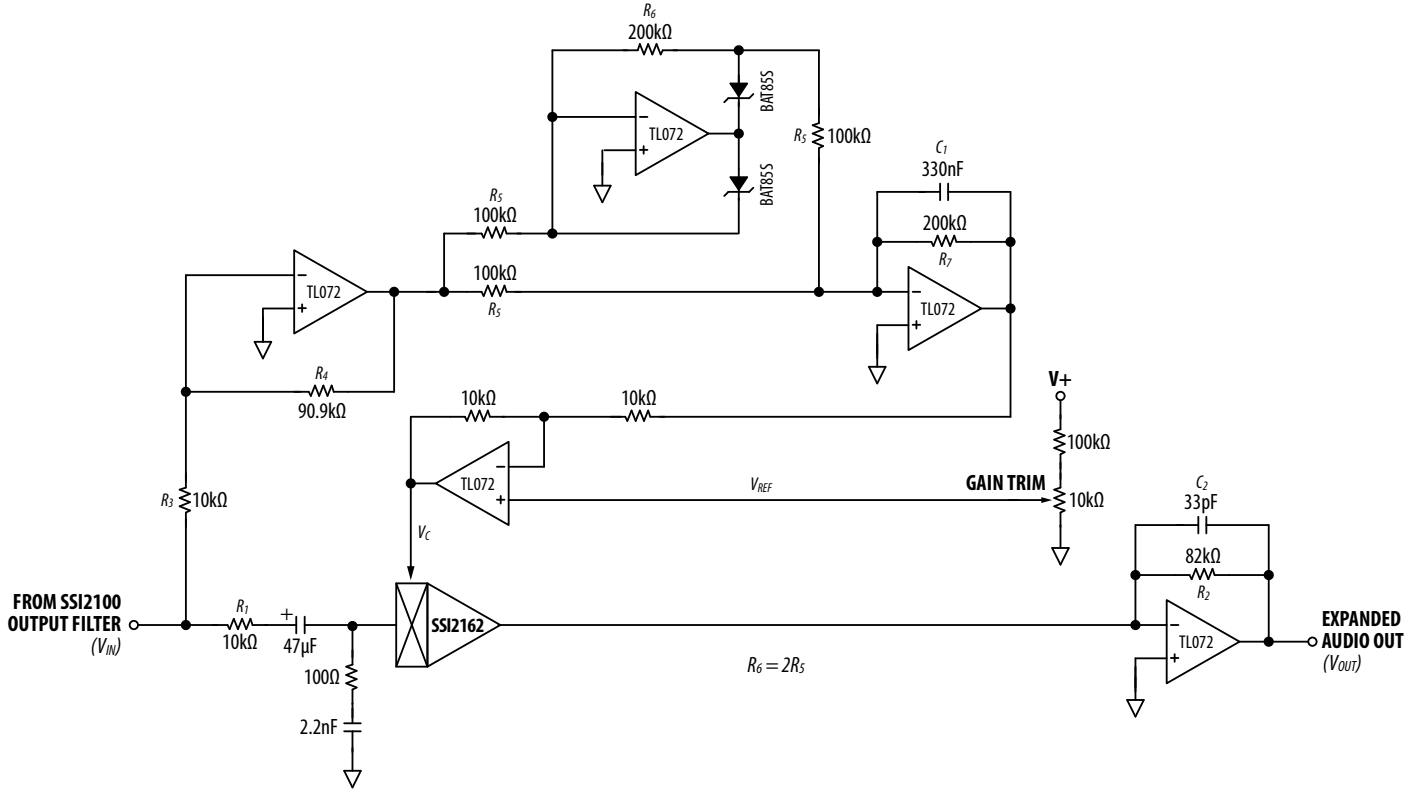


Figure 14: Expander

The audio path is very similar to the compressor: a VCA and corresponding op amp with some gain. However, at low signal levels (-64dBu and below) the output signal is attenuated by the VCA by 16dB down to -80dBu. As the input signal level increases the expander adds additional gain resulting in the output signal rising at a rate above that of the input. Referring back to the gain chart, the expander has an input range of 50dB and an output range of 100dB, meaning this is a 1:2 expander.

The sidechain circuit mirrors that of the compressor – same gain, same rectifier, same time constants – and it behaves in a similar way. The final op amp in the sidechain provides two functions. It inverts the sense of the sidechain control voltage (switching the polarity from compression to expansion), and adds an offset from the GAIN TRIM trimmer. This operates the second half of the SSI2162 to increase the signal level and so counteract the compressor.

Math behind the expander circuit is simpler than the compressor due to the feed-forward design: the output is only a function of input. Again, let's first consider the signal path. The output voltage, V_{OUT} , is:

$$V_{OUT} = V_{IN} \frac{R_2}{R_1} 10^{\frac{3}{2} V_C} \quad (13)$$

In the expander the control voltage V_C is an inverted and offset version of the sidechain signal as derived in the compressor:

$$V_C = 2V_{REF} - \frac{R_7}{R_5} \frac{R_4}{R_3} \frac{1}{\sqrt{2}} V_{IN} \quad (14)$$

Substituting (14) into (13) gives

$$V_{OUT} = V_{IN} \frac{R_2}{R_1} 10^{\left(-3V_{REF} + \frac{3R_7 R_4}{2R_5 R_3} \frac{1}{\sqrt{2}} V_{IN} \right)}$$

Bring out the constant term in the exponent, and then re-arranged into a simplified form:

$$V_{OUT} = V_{IN} \frac{R_2}{R_1} \frac{1}{10^{3V_{REF}}} 10^{\left(\frac{3R_7 R_4}{2R_5 R_3} \frac{1}{\sqrt{2}} V_{IN} \right)} \quad (15)$$

The resulting curve of equation 15 looks like the opposite of the compressor's:

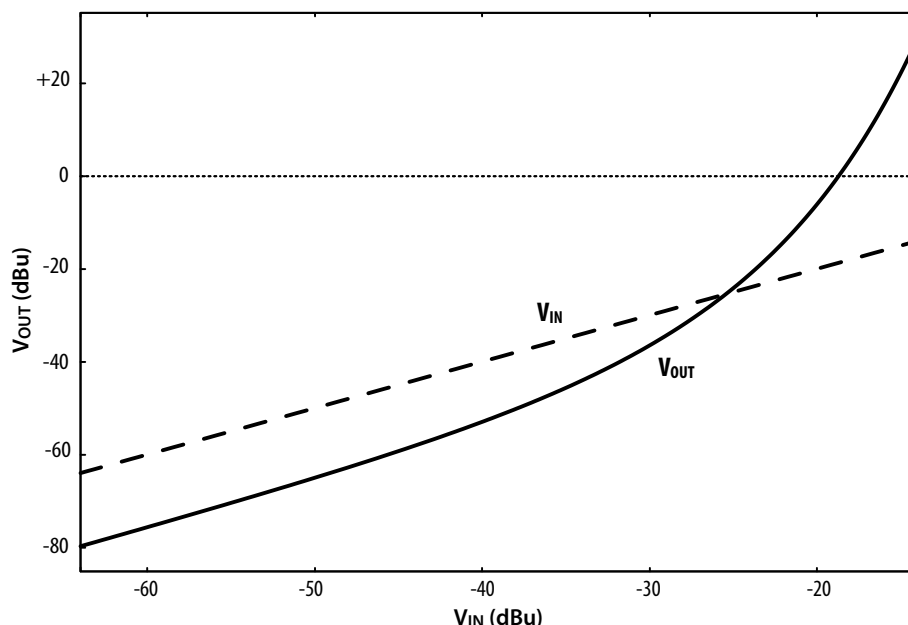


Figure 15: Expander Transfer Curve

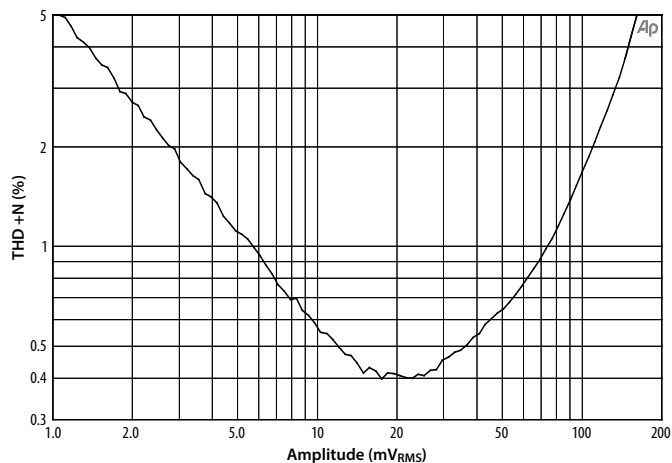
At very low signal levels the output (solid line) is below the input (dotted line). As the input signal increases the expander gain also increases. Again, at around -25dBu the expander is at unity gain, and then applies more gain to the signal. When combined with the compressor curve, the result is a near-perfect match where the output signal level follows the input signal level with very little difference over the range -80dBu to +20dBu.

The SSI2162 is an ideal choice for a compandor. It has two low-distortion and low-noise VCA channels, and exponential sensitivity of the control inputs means they operate in a very predictable volts-per-dB scale.

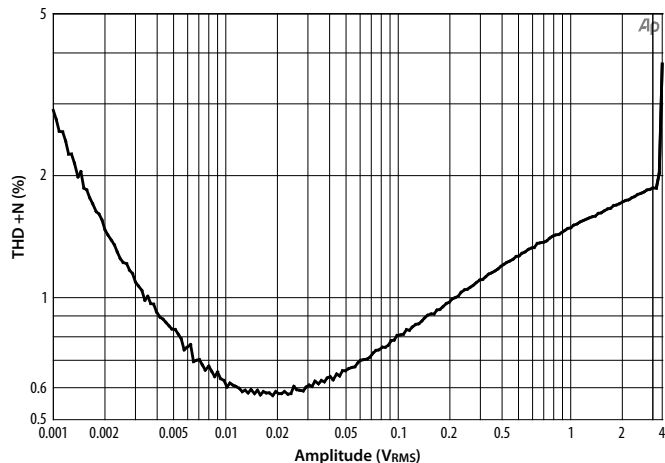
While the circuit as shown has been designed for SSI BBDs, it can also be applied to other signal range-limited applications; for example, wavefolders, analog recording/playback systems, and so on.

TYPICAL PERFORMANCE GRAPHS

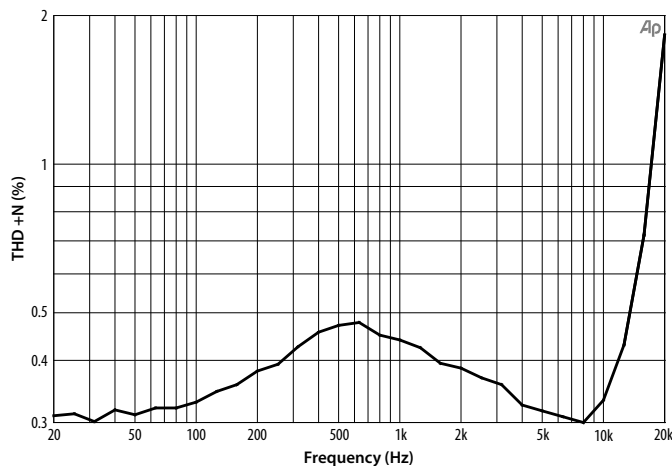
($V_+ = 5.0V$, $V_{DCB} = 3.2V$, $G_{CAP} = GND$, $f_{CLK} = 40kHz$, $V_{IN} = 20mV_{RMS}$, $f_{IN} = 1kHz$, $T_A = 25^\circ C$; using Figure 1 circuit unless otherwise noted)



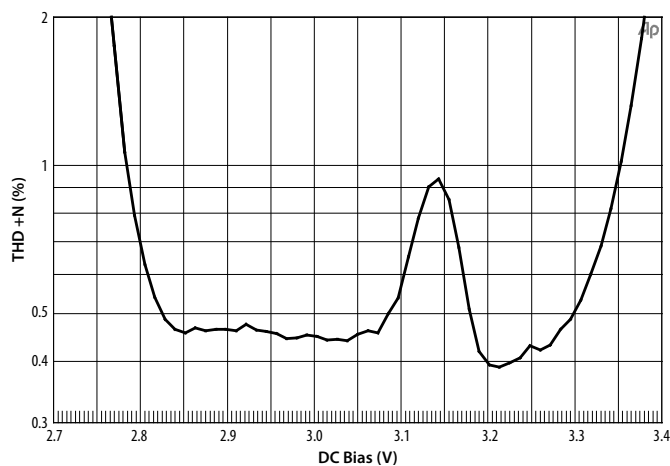
THD+N vs. Input Signal Amplitude - Compandor Out



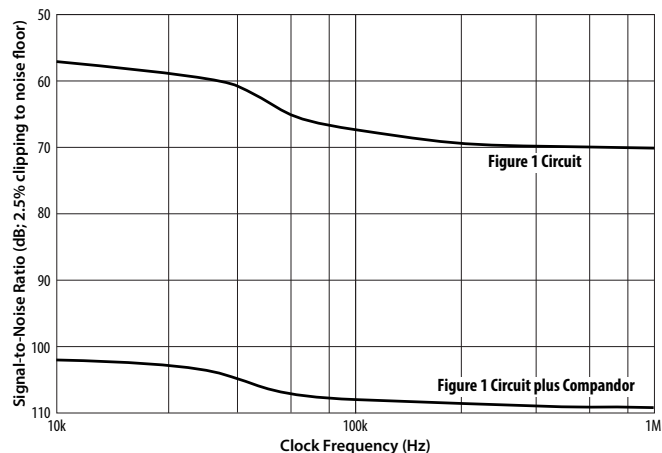
THD+N vs. Input Signal Amplitude - Compandor In



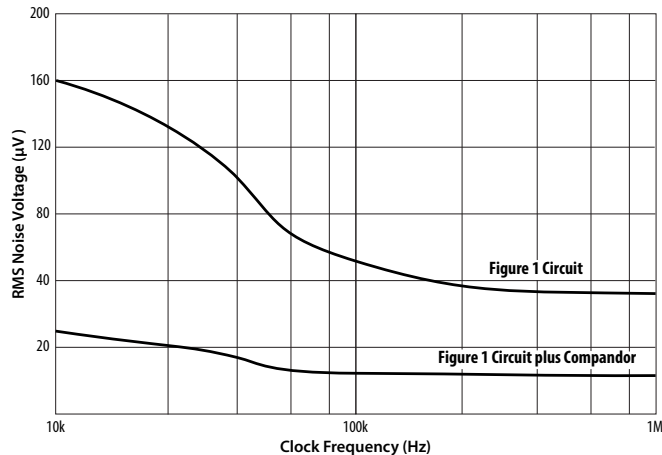
THD+N vs. Frequency (Hz)



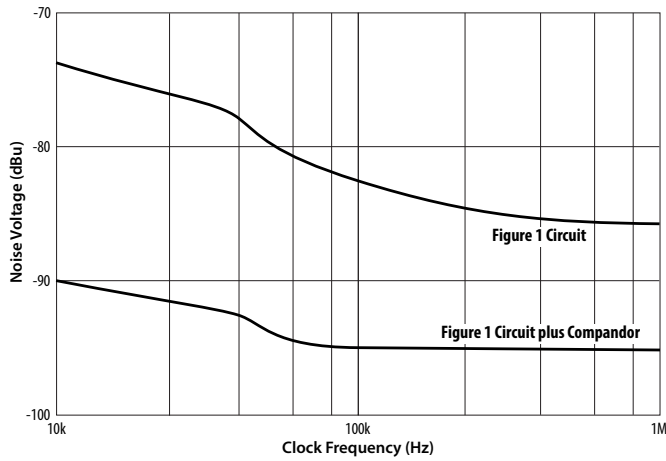
THD+N vs. DC Bias Voltage (V_{DCB})



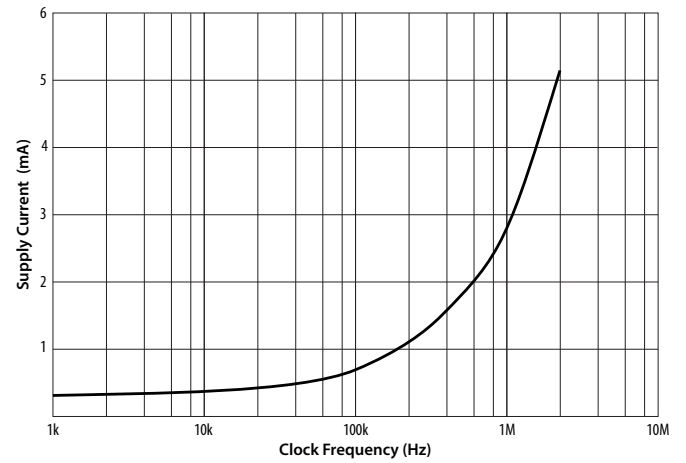
Signal to Noise Ratio vs. Clock Frequency



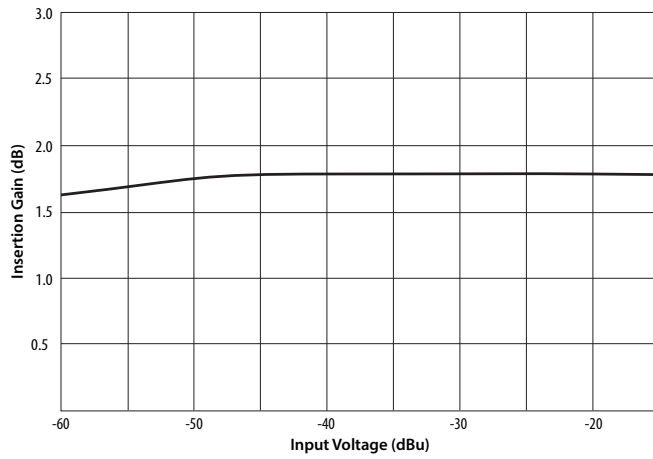
RMS Noise Voltage vs. Clock Frequency



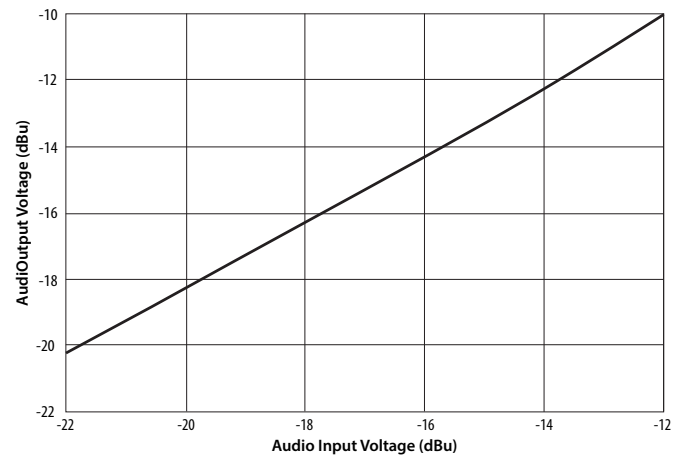
dBu Noise Voltage vs. Clock Frequency



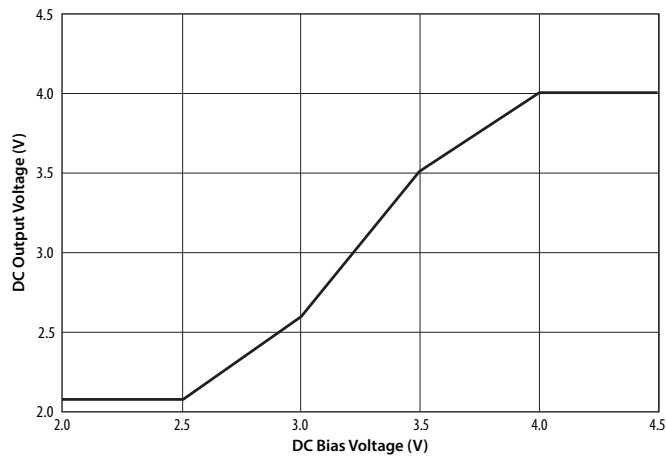
Supply Current vs. Clock Frequency



Insertion Gain vs. Input Voltage



Audio Input Voltage vs. Output Voltage



DC Bias Input Voltage (V_{DCB}) vs. Output Voltage